DATA MANAGEMENT CARPENTRY · EPISODE 9

Choosing a Data Storage Format

Understanding file extensions, open vs. proprietary architectures, discipline-specific standards, and format conversion hazards.

EPISODE SCOPE

Questions & Objectives

Core Questions

How do extensions dictate file structure?

What factors guide format selection?

Domain-specific vs cross-cutting formats?

What are the risks of converting formats?

Learning Objectives

Map file extensions to internal data layouts.

Evaluate portability, metadata, and scale tradeoffs.

Identify specialised data formats.

Analyse data fidelity and compression loss.

THE CORE CHALLENGE

The Wrong Format Can Break Your Data

The Software Default Trap

Researchers often save files in whatever software application happens to be open (e.g. saving spreadsheets as .xlsx or text as .docx).

> *"The format you choose dictates who can open your data, whether metadata survives, and if your data is readable in 20 years."*

Downstream Consequences

- Proprietary software obsolescence / lock-in
- Automated data corruption (e.g. gene names to dates)
- Loss of embedded instrumentation metadata
- Irreversible loss of precision via lossy conversions

FUNDAMENTALS

File Extensions & Encoding Architectures

Plain-Text Formats

Human-readable ASCII/UTF-8 characters. Can be inspected in any basic text editor.

Examples: .csv · .tsv · .txt · .json

Extremely portable and future-proof; however, larger footprint and slower for massive arrays.

Binary Formats

Machine-encoded byte sequences. Requires designated software parsers to interpret.

Examples: .xlsx · .h5 · .parquet · .nii

Compact, high-speed read/write performance; carries software dependency risk.

FORMAT PROFILES

Common Formats & Practical Trade-Offs

CSV (Comma-Separated Values)
Universal, open-source, long-term preservation standard. Cannot store multiple sheets, cell formatting, or formula logic.

Excel Workbook (.xlsx)
Convenient for multi-tab work and interactive formulas. Proprietary container; prone to auto-conversion bugs (e.g. gene names).

FWF (Fixed-Width Format)
Columns aligned by character count. Highly stable across legacy tools; wastes space on short values and risks truncating long fields.

JSON (Key-Value Trees)
Hierarchical key-value pairs ({"name": "Smith"}). Ideal for non-rectangular, nested, or evolving schema; verbose file sizes.

DISCUSSION 1 · 5 MINUTES

What to Consider When Choosing a Format?

Scenario: You are launching a 3-year research project generating thousands of data files.

Working in pairs: What criteria should guide which file format to adopt across your team?

- Who needs to open the data?
- How will metadata be embedded?
- Can tools read it in 10–20 years?

DOMAIN ARCHITECTURES

Discipline-Specific Research Formats

Neuroscience & Imaging: .nii

NIfTI embeds 3D/4D scan arrays with voxel dimensions and spatial orientation affine matrices.

Genomics & Sequencing: .fastq

Text structure binding DNA/RNA sequence base calls directly to PHRED quality scores.

Astronomy: .fits

Flexible Image Transport System combining multi-dimensional arrays with rich, self-documenting ASCII header blocks.

Geophysics: .asdf

Adaptable Seismic Data Format pairing waveform traces with station and event XML metadata.

EXECUTION SECURITY

Language-Specific Formats & Security Risks

Language-Bound Data Storage

- **Python:** .pkl (Pickle)
- **R:** .rda / .rdata
- **MATLAB:** .mat
- **Julia:** .jld2



Arbitrary Code Execution Hazard

Pickle files can execute arbitrary system code when unpickled by an unsuspecting user.

Not recommended for long-term storage or data exchange for this reason

SCALABLE SOLUTIONS

Cross-Discipline Formats: HDF5 & Parquet

HDF5 (.h5 / .hdf5)

"A file system within a single file."

Stores multi-dimensional numeric arrays alongside custom metadata attributes.

Adopted widely in climate models, electron microscopy, and deep learning weights.

Apache Parquet (.parquet)

Columnar Big Data Storage

Preserves strict data types and schema. Extremely fast for querying subsets of columns.

Ideal for massive tabular datasets that will be written once and queried many times.

EVALUATION MATRIX

The Format Evaluation Checklist

Open vs. Closed

Is the spec maintained by a public standard?

Self-Documentation

Can sensor settings and units live in the file?

Data Topology

Is data flat 2D, or multi-dimensional grid arrays?

Readability vs Size

Human inspectable text vs high-density binary.

DISCUSSION 2 · 3 MINUTES

Formats in Your Research Domain

Discuss with your neighbour:

- What primary formats do you generate or analyze daily?
- Are these formats open, or locked to proprietary instruments?
- What specific scientific challenges do these formats solve?

DATA FIDELITY

The Dangers of Format Conversion

Lossless: BMP & PNG

Mathematical Bit-Level Equality

Every pixel intensity is preserved exactly. Essential for automated segmentations, cell counts, and intensity quantifications.

Lossy: JPEG

Permanent Information Destruction

Discards high-frequency visual information to save space. Creates halos and blocky compression artifacts around edges.

PRACTICAL EXERCISE

Challenge: Comparing Image Formats

Inspect two files in your sample folder:

Penguin958.png · penguin958.jpg

1. Zoom into high-contrast cell boundaries at 400% in the jpg.
2. What visual distortions (halos, blockiness) appear?
3. What size are the files?

CHALLENGE FINDINGS

Solution: The Scientific Dangers of JPEG

Compression Artifacts

Zooming in reveals 8x8 pixel macroblocks and blurry rings around edges. Pixel intensity measurements are permanently corrupted.

Scripted Analysis Rule

Automated scripts misread artifacts as actual cells or miss faint signals completely.

Never use lossy formats (JPEG) for raw data intended for quantitative analysis.

STANDARDS GUIDANCE

Data Facility Recommendations

UK Data Service & Library of Congress

- **Tabular:** CSV, TSV, OpenDocument (.ods)
- **Images:** TIFF (uncompressed), PNG, SVG
- **Audio/Video:** FLAC, WAV, MP4 (MPEG-4)

Preservation Strategy

When finishing a project, export primary working files into open, non-proprietary archive formats so future researchers can open your data without commercial licenses.

SUMMARY

Episode 9 Key Points

File Extensions: Guide operating systems and parsers on how to decode raw bits.

Open Formats: Guarantee long-term access, transparency, and freedom from vendor lock-in.

Domain Standards: NIfTI and FASTQ meet specialized needs; HDF5 spans cross-cutting arrays.

Avoid Lossy Conversions: Formats like JPEG permanently discard data and ruin scientific fidelity.

The Invariant: Integrate metadata directly within file headers to ensure self-documenting research.

COURSE COMPLETION

Data Management Carpentry Complete

You have journeyed from a chaotic inherited project folder to a reproducible, documented, secure, and standardized scientific data workflow.

Findable · Accessible · Interoperable · Reusable